

■ Polygon

`Polygon[{ $pt_1$ ,  $pt_2$ , ...}]` is a graphics primitive that represents a filled polygon.

`Polygon` can be used in both `Graphics` and `Graphics3D` (two- and three-dimensional graphics). ■ The positions of points can be specified either in absolute coordinates as $\{x, y\}$ or $\{x, y, z\}$, or in scaled coordinates as `Scaled[{ $x, y$ }]` or `Scaled[{ $x, y, z$ }]`. ■ The boundary of the polygon is formed by joining the last point you specify to the first one. ■ In two dimensions, self-intersecting polygons are allowed. ■ In three dimensions, the two-dimensional convex hull of the points is formed. For a planar polygon that does not intersect itself, the polygon drawn will be exactly as you specified it. ■ You can use the graphics primitives `GrayLevel` or `RGBColor` to specify how polygons should be filled. ■ In three dimensions, the shading can be produced from simulated illumination. ■ In three-dimensional graphics, polygons are considered to have both a front and a back face. The sense of a polygon is defined in terms of its first three vertices. When taken in order, these vertices go in a *counter-clockwise* direction when viewed from the *front*. (The frontward normal is thus obtained from a *right hand* rule.) ■ You can use `FaceForm` to specify gray level or color for the the front and back faces of polygons. ■ In three-dimensional graphics, intersections between polygons are shown as lines, with forms specified by the graphics primitive `EdgeForm`. ■ See pages 142 and 149. ■ See also: `CellArray`, `Rectangle`.